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1 August 1984
SUPERSEDING
MIL-R-46710A
Amendment 2
11 December 1964

MILITARY SPECIFICATION

RECOVERY VEHICLE, FULL-TRACKED: LIGHT, ARMORED, M578; PROCESSING FOR STORAGE AND SHIPMENT OF

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the processing of the M578 Armored, Light, Full-Tracked, Recovery Vehicle, 8736340, for storage and shipment.

1.2 Classification. Processing shall be of the following levels as specified (see 6.2).

Level A - Maximum military protection.

Level A is the processing required for the protection of vehicles during shipment, handling, and storage exceeding 90 days from date of actual processing. This level does not provide for driveaway capability. It does provide for domestic or overseas shipment, including open deck loading.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

NO DELIVERABLE DATA REQUIRED

AREA PACK

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Level B - Minimum military protection.

Level B is the limited processing required for the protection of vehicles during shipment, handling, and storage not to exceed 90 days from date of actual processing. This level provides for driveaway capability when specified, and domestic or overseas shipment (excluding open deck loading).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS
FEDERAL

A-A-883	- Tape, Pressure Sensitive Adhesive, Masking.
O-A-548	- Antifreeze/Coolant Engine: Ethylene Glycol, Inhibited, Concentrated.
O-I-490	- Inhibitor, Corrosion, Liquid Cooling System.
O-S-576	- Sodium Bicarbonate, Technical.
O-S-801	- Sulfuric Acid, Electrolyte: for Storage Batteries.
P-D-220	- Detergent, General Purpose.
GG-T-250	- Tester, Antifreeze Solutions.
NN-P-530	- Plywood, Flat Panel.
QQ-S-781	- Strapping, Steel, and Seals.
RR-W-360	- Wire, Fabric, Industrial.
TT-E-529	- Enamel, Alkyd, Semigloss.
TT-V-121	- Varnish, Spar, Water Resisting.
UU-P-268	- Paper, Kraft, Wrapping.
UU-T-81	- Tags, Shipping and Stock.
VV-L-800	- Lubricating Oil, General Purpose, Preservative (Water-Displacing, Low Temperature).
MMM-A-178	- Adhesive, Paper Label, Water-resistant.
PPP-B-621	- Box, Wood, Nailed and Lock Corner.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-B-1055	- Barrier Material, Waterproof Flexible.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-T-60	- Tape, Packaging, Waterproof.

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MIL-P-116	- Preservation, Methods of.
MIL-B-117	- Bags, Sleeves and Tubing - Interior Packaging.
MIL-C-450	- Coating Compound, Bituminous, Solvent Type, Black (for Ammunition).
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-C-5501/7	- Caps and Plugs, Protective, Dust and Moisture Seal (Cap-Plug, General Purpose).
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-A-11755	- Antifreeze, Arctic-Type.
MIL-B-12841	- Basic Issue Items for Military Vehicles, Carriages and Equipment, Preparation for Shipment and Storage of.
MIL-P-13983	- Paint, Temporary, Lusterless, Gasoline Removable.
MIL-P-14232	- Parts, Equipment and Tools for Army Materiel, Packaging of.
MIL-L-14486	- Lacquer, Vinyl Resin, Semi-Gloss.
MIL-C-16173	- Corrosion Preventive Compound, Solvent Cutback, Cold-Application.
MIL-D-16791	- Detergent, General Purpose (Liquid, Nonionic).
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-I-22110	- Inhibitors, Corrosion, Volatile, Crystalline.
MIL-P-46002	- Preservative Oil, Contact and Volatile Corrosion-Inhibited.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-T-50036	- Talc, Technical, T1 and T3.
MIL-D-81298	- Dye, Liquid for the Detection of Leaks in Aircraft Fuel Systems.

STANDARDS
MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-647	- Packaging Standards, Preparation and Use of.
MS15795	- Washer, Flat-Metal Round, General Purpose.

2.1.2 Other Government documents, drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

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DRAWINGS
ARMY

10881915	- Strut.
10881916	- Kit, OVE Rack.
10881917	- Bracket.
10914995	- Kit, Vehicle Closure.
10919932	- Bracket.
10919933	- Bracket.
10919967	- Stringer.
10919968	- Support.
10919969	- End.
10919979	- Hull Modification.

PUBLICATIONS
PACKAGING DATA SHEET

MS-35000-3	- Battery, Dry Charge, Storage, Type 6TN.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASSOCIATION OF AMERICAN RAILROADS PUBLICATIONS

Section No. 1	- General Rules Governing Loading of Commodities on Open Top Cars.
Section No. 6	- Rules Governing the Loading of Department of Defense Material on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren, Chicago, Illinois 60605.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), one of the first 10 production processed vehicles shall be subjected to the inspection specified in 4.2. Approval of this vehicle shall not relieve the contractor of his obligation to process all vehicles in accordance with this specification. Unless otherwise specified by the acquisition activity, any change to materials or design after approval shall require additional vehicles be inspected as specified in 4.2. Government representative shall select the vehicle to be inspected.

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3.2 Materials. Materials shall be as specified herein and on the standards, specificatons, and drawings referenced herein (see 4.5.1).

3.3 Processing. Processing levels A and B shall be as specified below, with level B processing modified where specified.

3.3.1 Preparation prior to processing. Except as otherwise specified herein, and to the maximum extent consistent with production efficiency, economy, and safe storage and shipment, the vehicle shall be prepared for storage and shipment in a completely assembled condition after test runs, and completion and approval of all necessary repairs. Specified equipment shall be installed and all adjustments made so that the vehicle may be operated, shipped, and placed into service with a minimum of delay.

3.3.1.1 Processing records. Records of vehicle processing shall be maintained and shall be readily available for review by Government representatives.

3.3.2 Disassembly. Projecting parts whose removal will accomplish desired cube reduction, and parts susceptible to damage and pilferage, shall be removed from the vehicle. Removed bolts, nuts, screws, pins, and washers shall be placed in one of the mating parts and secured. Removed parts shall be packaged, identified, and stowed securely within the vehicle.

3.3.2.1 Matchmarking. Parts removed from the vehicle shall be matchmarked when necessary to facilitate reassembly. Matchmarking information shall be put on cloth shipping tags conforming to type A of UU-T-81 or on metal tags using waterproof ink or paint, and attached to mating parts. The marked cloth shipping tags shall be waterproofed with varnish conforming to TT-V-121 or adhesive conforming to MMM-A-178.

3.3.3 Record forms. Two copies of DD Form 1397 shall be provided. Information on forms shall include preservation accomplished and depreservation instructions. The Equipment Log Book Binder and one copy of DD Form 1397 (see 6.4) shall be placed in a bag conforming to type I, class B, style 2, 6 mil of MIL-B-117; the bag shall be closed by heat sealing and securely attached in driver compartment of the vehicle. The other copy of DD Form 1397 shall be water-proofed with adhesive conforming to MMM-A-178 or placed in a bag conforming to type I, class B, style 2 of MIL-B-117. The bag shall be closed by heat sealing and securely attached in a conspicuous location on the exterior of the vehicle.

3.3.4 Cleaning and drying (see 4.5.2.1).

3.3.4.1 Vehicle interior. Interior surfaces of the vehicle shall be cleaned with a solution of detergent conforming to P-D-220 or type I of MIL-D-16791 and water. The solution temperature shall not exceed 210 degrees Fahrenheit (°F) and the pressure shall not exceed 5 pounds per square inch (psi) measured 4 inches from the nozzle. After cleaning, the surfaces shall be rinsed with clean water and dried. Care shall be taken during cleaning and rinsing operations to assure that no solution or water enters the

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instruments, connections, or other components susceptible to water damage. The solution or water shall not accumulate and remain in cavities that cannot be drained. Vehicles with decals, markers, and straps and floor plates installed shall only be hand cleaned with solution of P-D-220 or type I of MIL-D-16791 and water to prevent damage to these components. The cleaned surfaces shall be hand rinsed and dried.

3.3.4.1.1 Cleaning and drying of battery supports and retainers. The battery supports and retainers shall be cleaned with a solution of one-half pound of sodium bicarbonate conforming to O-S-576 per gallon of water. After cleaning, the surfaces shall be flushed with clean water and thoroughly dried. Dried surfaces shall then be preserved in accordance with 3.3.5.2.

3.3.4.1.2 Cleaning and drying of backrests, seats and crash pads. Backrests, seat cushions and crash pads shall be cleaned with a solution of detergent conforming to P-D-220 or type I of MIL-D-16791, in warm water. After cleaning, cushions shall be wiped with cloth saturated with clean water to remove cleaning solution. Care shall be taken not to saturate the cushions with cleaning solution or water. After rinsing, cushions shall be dried and protected in accordance with 3.3.16.

3.3.4.1.3 Fuel filter cases. The fuel filter cases, except when they are new, clean, and prior to engine preservation (see 3.3.5.5), shall be drained, removed and cleaned in accordance with process C-1 of MIL-P-116. After drying by any applicable procedure specified therein the fuel filter cases shall then be reinstalled using new gaskets.

3.3.4.2 Vehicle exterior. The vehicle exterior shall be cleaned using a solution of detergent conforming to P-D-220 or type I of MIL-D-16791, in warm water or steam. Cleaning shall remove all foreign matter. After cleaning, surfaces shall be rinsed with clean water or steam and thoroughly dried. Care shall be taken to avoid entry of water or steam into driver or engine compartments.

3.3.5 Preservation.

3.3.5.1 Relubrication. If the vehicle has been operated more than 50 miles since lubrication or after the vehicle has been cleaned in accordance with 3.3.4.2, the vehicle shall be relubricated using materials conforming to the drawing, specification, or lubrication order applicable to the vehicle. All exposed oil can points such as levers, locking levers, locking bars, locking pins, pintle pins, hinge pins, hinges, strikers, wing nuts, door locks, hand-operated locking knobs, latches, linkage, threaded ends of yokes, and related clevis pins shall be coated with lubricant conforming to VV-L-800. Excess lubricant shall be removed after coating.

3.3.5.2 Preservation of battery supports and retainers. The top battery supports and retainers shall be preserved with compound conforming to composition G, type I or II of MIL-C-450.

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3.3.5.3 Transmission, final drives and auxiliary drive gear box. Transmission, final drives and auxiliary drive gear box shall be filled to operating level with lubricant in accordance lubrication orders. If preservative oil is used fill with Grade 10W, 30 or 50, as applicable, in accordance with MIL-L-21260. Operating lubricant and preservative oil shall not be mixed. DD Form 1397 shall be annotated with the grade of lubricant or preservative oil used (see 3.3.3).

3.3.5.4 Engine crankcase preservation. The crankcase shall be filled to operating level with lubricating oil conforming to MIL-L-21260 of the seasonal grade specified in the applicable drawing, specification, or lubrication order. DD Form 1397 shall be annotated with the type and grade of lubricant used.

3.3.5.5 Engine preservation. For level A processing, the engine preservation shall be an uninterrupted sequence in accordance with 3.3.5.5.2 through 3.3.5.5.5.

3.3.5.5.1 Level B. For level B processing, the engine shall be preserved in accordance with 3.3.5.5.4 and 3.3.5.5.5 only. Preservation through the fuel system (see 3.3.5.5.2) and through the combustion chamber (see 3.3.5.5.3) shall not be required.

3.3.5.5.2 Preservation through fuel system. An auxiliary fuel container, equipped with a fuel line shall be filled with a sufficient amount of preservative oil conforming to grade 1 of MIL-P-46002 to which an oil soluble red dye conforming to MIL-D-81298 has been added, in a concentration sufficient to impart a marked coloring to the oil, to operate the engine as prescribed below. The container shall be arranged to provide adequate pressure to allow a proper supply of the preservative oil into the combustion chambers. The fuel line shall be disconnected at the most convenient point nearest to the vehicle fuel tank. The line from the auxiliary fuel container shall be connected to the fuel-to-engine line at the point of disconnect. The vehicle fuel return line shall be disconnected at the quick disconnect coupling. A transparent plastic line shall be connected to the disconnected vehicle fuel return line and the other end inserted into a recovery container to collect the returned diesel fuel. The fuel valve on the auxiliary fuel container shall be turned to the "ON" position. The engine shall be started and operated not over 1200 revolutions per minute (rpm) until observed fuel return is purged of diesel fuel. The plastic fuel return line shall then be switched to an auxiliary container to recover the undiluted preservative oil. The engine shall be operated on the preservative oil for not more than one minute after the oil appears through the transparent plastic line. The engine shall then be accelerated to three-fourths of the maximum speed for approximately 15 seconds and then shut-off.

3.3.5.5.3 Preservation through combustion chamber. After preservation through the fuel system, with the container still connected (see 3.3.5.5.2), the auxiliary fuel tank shall be refilled with enough preservative oil conforming to grade 1 of MIL-P-46002 to assure a proper supply. The engine shall be cooled to a cylinder head temperature of not more than 100°F, before engine preservation is continued. Induced air currents may be used to accelerate engine cooling. Engine rocker arm covers shall be removed to gain access to valve mechanisms. One valve of each cylinder shall be fixed in an

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open position by inserting a 0.040 inch (maximum) spacer between the valve and the rocker arm to release compression. The fuel valve on the auxiliary fuel container shall be turned to the "ON" position. The accelerator shall be depressed to full fuel flow and the engine shall then be cranked for a period of not less than 30 seconds and not more than 40 seconds. After cranking, the spacers shall be removed and rocker arm covers reinstalled. The auxiliary fuel container shall be removed and the vehicle fuel and return lines reconnected.

3.3.5.5.4 Preservation through air intake and exhaust systems. After preservation through the fuel system (see 3.3.5.5.2) and combustion chamber (see 3.3.5.5.3), the engine crankcase breathers shall be sealed with plastic caps conforming to table I of MIL-C-5501/7, or tape conforming to type III, class 1 of PPP-T-60. Twenty grams of volatile corrosion inhibitor compound(VCI) conforming to type I of MIL-I-22110 shall be fogged into the exhaust outlet opening. The opening shall then be sealed with tape conforming to type III, class 1 of PPP-T-60. The hose shall be disconnected from the air duct in the crew compartment and 15 grams of VCI shall be fogged into the air duct opening. The opening shall then be sealed with a plastic plug or tape conforming to type III, class 1 of PPP-T-60. A red warning tag, bearing the information "ENGINE PRESERVED WITH VCI AND OIL - DO NOT CRANK" and "BEFORE CRANKING, REMOVE CAPS, TAPE AND PLUG, AND REINSTALL HOSE" shall be placed in a conspicuous location within the driver's compartment. DD Form 1397 shall be annotated to show the engine is preserved with VCI and oil and openings are sealed.

3.3.5.5.5 Preservation through dipstick shroud opening. After the preservation specified in 3.3.5.5.2, 3.3.5.5.3, and 3.3.5.5.4, the dipstick shall be removed and six ounces of lubricating oil conforming to grade 1 of MIL-P-46002 shall be sprayed in atomized form into the crankcase through the dipstick shroud opening. An extension nozzle of sufficient length to allow the spraying nozzle to be within the crankcase shall be used. The nozzle shall not be submerged in the oil within the crankcase. The dipstick shall be reinstalled. All openings leading to the interior of the engine, including the dipstick shroud opening and oil filler cap, shall be sealed with tape conforming to type III, class 1 of PPP-T-60.

3.3.5.6 Fuel tank. After engine preservation, the fuel tank shall be drained to the extent possible. The fuel tank shall be atomized sprayed with lubricating oil conforming to grade 30 of MIL-L-21260 in a manner to assure complete coverage of all interior surfaces. The fuel tank cap and filler screen shall be removed and coated with the grade 30 preservative oil and reinstalled.

3.3.5.6.1 Level B. Unless otherwise specified (see 6.2), vehicles processed according to the level B shall be shipped without draining residual fuel from the fuel tanks.

3.3.6 Cooling system. The cooling system shall be protected by one of the following procedures (see 6.2):

a. For shipment and storage in areas where temperatures drop below minus 40°F, the systems shall be protected as specified in 3.3.6.3.

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b. For shipment and storage within the bounds of 30 degrees north latitude and 20 degrees south latitude, except continental United States, systems shall be protected as specified in 3.3.6.2.
c. For all other shipments, cooling systems shall be protected as specified in 3.3.6.1.

NOTE: DD Form 1397 (see 3.3.3) shall be completed to indicate coolant used.

3.3.6.1 Water and antifreeze procedure. The cooling system shall be filled to operating level with a clean solution consisting of equal parts by volume of antifreeze (ethylene glycol) conforming to type I of O-A-548 and water. The engine shall be operated until a temperature has been reached that causes the thermostat to open to assure complete mixing and even distribution of the antifreeze solution. A warning tag bearing the information, "COOLING SYSTEM FILLED WITH WATER AND ANTIFREEZE SOLUTION (ETHYLENE GLYCOL) IN EQUAL PARTS BY VOLUME - DO NOT DRAIN", shall be securely attached to radiator filler neck.

3.3.6.2 Water and corrosion inhibitor procedure. The cooling system shall be filled with clear water up to the radiator upper tank. A corrosion inhibitor conforming to O-I-490 shall be added in the proportion of 5 ounces of the inhibitor for each 10 quarts of water. The inhibitor shall be dissolved in 2 quarts of warm water and poured into the radiator while the engine is idling. More water shall be added if necessary, to fill the radiator to operating level. A warning tag bearing the information, "COOLING SYSTEM DOES NOT CONTAIN ANTIFREEZE - FILLED WITH WATER AND INHIBITOR", shall be securely attached to radiator filler neck.

3.3.6.3 Antifreeze compound procedure. The cooling system shall be filled to the operating level with antifreeze compound conforming to MIL-A-11755. The compound shall be used without dilution. A warning tag bearing the information, "COOLING SYSTEM FILLED WITH ANTIFREEZE (ARCTIC-TYPE) - DO NOT DRAIN", shall be securely attached to radiator filler neck.

3.3.7 Batteries, cables, and electrolyte.

3.3.7.1 Dry charge batteries and cables. The positive battery cables shall be taped to the battery-to-ground cable with tape conforming to type III, class 1 of PPP-T-60. The battery supports, retainers, and related hardware shall be grouped together and secured to the battery cables with the same type and class of tape. The dry charge batteries shall be packaged and packed in accordance with level A requirements of Packaging Data Sheet MS-35000-3. Instructions for activation of batteries shall be packed with the battery. The dry charge batteries shall be stowed with OVE and secured independently in a manner to prevent movement and damage in transit.

3.3.7.2 Electrolyte.

3.3.7.2.1 Packaging. The electrolyte shall be packaged and marked in accordance with the requirements prescribed for class 2, 3, or 4, as applicable, of O-S-801. The packaged electrolyte shall be packed in accordance with 3.3.7.2.2 in lieu of the packing requirements specified for class 2, 3, or 4 container in O-S-801.

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3.3.7.2.2 Packing. The packing of the required number of unit packages of electrolyte shall be in crate-type structures constructed with solid ends. The ends shall be of nominal 1 inch lumber. Four slats plywood, interior with exterior glue, conforming to group B of NN-P-530 species group 1, 2, or 3 only, having a length to fit, 4 inches wide by 1/4 inch thick shall form the sides. One slat is required per side. Each slat shall be secured to the solid crate ends with two 3d sinker or cooler cement coated nails. The nails shall be placed one inch from the slat side, two inches on center. The crate shall be banded with 3/4-inch strapping conforming to type I, class B of QQ-S-781. The strapping shall run lengthwise, with the straps placed at the approximate center of slats, and crisscrossed at right angles over the solid ends. Corner protectors shall be used under the strapping. If markings on electrolyte unit packages are not readable through the slats, the same marking information shall be applied on the exterior surfaces of the crate. Arrows shall be applied to each crate to indicate correct stacking. The electrolyte crate shall be stowed with OVE and secured independently in a manner to prevent movement and damage in transit. The electrolyte shall be stowed in a manner permitting easy removal at ports when special stowing is required by maritime regulations.

3.3.8 Traverse ring gear. All exposed unpainted surfaces of the traverse ring gear shall be coated with grease conforming to MIL-G-10924.

3.3.9 Spade. The spade shall be extended from the retracted position. Approximately one inch of exposed surface of hydraulic piston rods shall be coated with grease conforming to MIL-G-10924. The spade shall then be returned slowly to full retracted position.

3.3.10 Winch and derrick assembly. The winch gearcase and other related gear driven units shall contain the applicable grade of gear lubricant conforming to MIL-L-2105, filled to the operating level. The wire cable shall be unreeled and all surfaces shall be coated with preservative conforming to grade 1 of MIL-C-16173. While the cable is being rewound, any damage to applied preservative coating shall be repaired by application of additional preservative of the same type and grade to damaged areas. All exposed unpainted metal surfaces of cable drums, sheaves, snatch blocks, boom block, crane or derrick boom, controls, and linkage shall be coated with preservative conforming to grade 1 of MIL-C-16173. All moving, mating parts shall be coated with grease conforming to MIL-G-10924. The hydraulic system shall contain lubricating oil, conforming to MIL-L-46167 filled to operating level. When operating surfaces of the hydraulic piston rod or ram are exposed they shall be coated with grease conforming to MIL-G-10924.

3.3.11 Machinegun mount. The machinegun mount shall be removed, cleaned, coated with grease conforming to MIL-G-10924, then overwrapped in barrier material conforming to PPP-B-1055. The barrier material shall be secured with tape conforming to type II, class 1 of PPP-T-60, identified and stowed inside the cab and secured in a manner to prevent movement during shipment.

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3.3.12 Miscellaneous securement. The following items shall be secured in their normal stowed position on the vehicle as identified. When necessary to provide secure stowage, strapping conforming to class B of QQ-S-781 shall be used. The securement shall be accomplished without increasing vehicle cube.

a. Portable fire extinguisher on left interior of cab.
b. Towing bar on right fender.
c. Snatch block on left front of cab.
d. Snatch block in tray on bottom of boom.
e. Two towing cables, one on each side of boom.
f. Lifting chain in exterior cab stowage compartment.
g. Two 4 foot chains in exterior cab stowage compartment.
h. Two 6 foot chains in exterior cab stowage compartment.
i. Tow chain in exterior cab stowage compartment.
j. Acetylene cylinder on cab exterior.
k. Oxygen cylinder on cab exterior.
l. Six track shoes on deck.

3.3.13 Vision blocks. Vision blocks shall be removed, wrapped in 2 layers of cushioning material conforming to type I, class B of PPP-C-843 then overwrapped in paper, having a minimum basis weight of 60 pounds, conforming to grade B of UU-P-268. The paper shall be secured with tape conforming to type I of A-A-883. The wrapped vision blocks shall be placed in a box conforming to PPP-B-636, identified, stowed, and secured inside cab in a manner to prevent movement during shipment.

3.3.14 Cloth items. Cloth items such as canvas shall be thoroughly dried, folded or rolled in a manner to avoid damage, packaged in accordance with method IC-5 of MIL-P-116, and packed in a nailed wood box conforming to PPP-B-621. The box shall be identified and stowed with the Basic Issue Items (BII).

3.3.15 Hydraulic systems. Hydraulic systems shall be filled to operating level with lubricating oil conforming to MIL-L-46167.

3.3.16 Backrests, seats, and crash pads. Immediately after drying (see 3.3.4.1.2), cushioned portions of backrests, seats, and crash pads shall be covered with paper, having a minimum basis weight of 60 pounds, conforming to grade B of UU-P-268. The paper shall be secured with tape conforming to type I of A-A-883.

3.3.17 Hatches and doors. Rubber seals around hatches and doors shall be coated with powered talc conforming to MIL-T-50036. For shipment, hatches and doors shall be closed and locked, except that the driver's hatch shall be secured with an approved railroad type seal placed through the locking lugs. For storage, the driver's hatch shall be secured in the open position for ventilation.

3.3.17.1 Level B. For level B processing, coating rubber seals with talc shall not be required.

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3.3.18 Fire extinguishers. Fire extinguishers shall have a minimum of 90 percent of rated full charge. All seals shall be intact. DA form 253 shall be completed and securely attached to each extinguisher (see 6.3).

3.3.19 Miscellaneous preservation. Except as otherwise specified herein, all exposed, unpainted, machined-metal surfaces on the interior of the vehicle shall be coated with preservative conforming to grade 4 of MIL-C-16173. All exposed, unpainted machined-metal surfaces on the exterior of the vehicle shall be coated with preservative conforming to grade 1 of MIL-C-16173.

3.3.20 Basic issue item. Except as otherwise specified herein, all spare parts, tools, equipment and other unit packages shall be packaged level A in accordance with MIL-STD-647, MIL-P-14232, or the applicable packaging data sheet. The packaged items shall be packed level A and marked in accordance with MIL-B-12841. The pack shall be identified to the pertinent vehicle by serial number and, except during shipment, shall be stored inside buildings.

3.3.21 Vehicle closure. The vehicle shall be provided with a covered wagon-style closure in accordance with drawing 10914995. The closure shall be fabricated, assembled, and installed in accordance with the appendix and drawings. All sharp corners of the framework and bows where cloth cover will make contact, such as corners of the base of the frame, shall be cushioned with at least two thicknesses of cushioning material conforming to PPP-C-843 and secured in place with tape conforming to A-A-883. To lift the vehicle for loading, the cover shall be rolled away from the front and rear bows to expose the vehicle lifting eyes.

3.3.21.1 Ventilation. The turret well access door shall be secured in the open position with wire. The transmission and radiator drain (left side) access plates and gaskets on the underside of vehicle shall be removed for ventilation. Unpainted metal surfaces of access plates, and surfaces exposed by removal of these items shall be coated with preservative conforming to grade 1 of MIL-C-16173. Plates and gaskets shall be placed in a box conforming to PPP-B-636, identified, and stowed within cargo compartment in a manner to prevent movement in transit. The following information shall be stenciled on the exterior of the vehicle "REMOVE SCREENS, INSTALL ACCESS PLATES AND GASKETS, AND CLOSE TURRET WELL ACCESS DOOR BEFORE VEHICLE OPERATION." The stenciling color shall be white or yellow, using paint conforming to MIL-P-13983. Stenciling shall be in characters at least 3/4 inch high.

3.3.21.1.1 Screens. Screens (see figure 1) constructed of wire cloth, 0.047 inch diameter, 4 by 4 mesh, conforming ot type I, class 1 of RR-W-360 shall be installed in the transmission and radiator drain access plate openings (see figure 2). Another screen shall be placed over the turret well opening and secured with tape conforming to type III, class 1 of PPP-T-60, from the inside of the vehicle.

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3.3.22 BII containers.

3.3.22.1 Domestic shipment. Unless otherwise specified (see 6.2), for domestic shipment BII containers shall be placed on the floor of the railroad car and blocked, braced, and anchored to prevent movement in transit. Corner protectors shall be used under the strapping.

3.3.22.2 Oversea shipment. For oversea shipment, BII containers shall be placed on vehicle equipment rack (see 3.3.22.2.1) in a position which will not increase the overall cube of the vehicle. Each large wood box, such as those containing manufacturer and depot BII, shall be secured with two 1 1/4-inch stappings in each direction. Smaller BII boxes shall be secured to larger BII boxes or racks with 3/4-inch strapping. The strapping shall conform to type I, class B of QQ-S-781. Wood blocking shall be used as necessary to reinforce racks. Corner protectors shall be used under strapping.

3.3.22.2.1 BII rack. The vehicle shall be provided with an BII rack constructed, and assembled in accordance with 10881916 and mounted in accordance with 10919979. The BII rack shall be installed as follows:

a. Attach strut 10881915 to hull using screws and lockwashers.
b. Attach brackets 10881917 to hull using screws and lockwashers.
c. Fit ends 10919969 over ends of stringers 10919967 and secure with lag bolts, lockwashers, and plain washers.
d. Attach supports 10919968 to stringers 10919967 using bolts, washers, and nuts.
e. Secure assembled rack to brackets 10881917 and strut 10881915 using bolts, washers, and nuts.

3.3.23 Tow hooks. Tow hooks and related hardware shall be removed for shipment, placed in a box conforming to type I, class 2, grade 3 of PPP-B-636, sealed, identified, then stowed inside the vehicle.

3.3.24 Loading. Loading of vehicles on open top railroad cars shall be in accordance with the applicable requirements of Section 1, General Rules Governing the Loading of Commodities on Open Top Cars and figure 87 or 87A, Section 6, Rules Governing the Loading of Department of Defense Material on Open Top Cars, publication of the Association of American Railroads (see 2.2). The parking brake shall be set.

3.3.25 Reprocessing engine after loading. If the engine is operated in connection with loading, or moving the vehicle to a loading area, the engine shall be reprocessed as specified in 3.3.5.5. An auxiliary fuel tank shall supply the fuel. If installed, the vehicle cover shall be rolled clear of engine intake and exhaust (see 3.3.25.1) to provide air circulation and to prevent damage to the cover. After reprocessing of the engine, the vehicle cover shall be restored to its original position.

3.3.25.1 Instructions for rolling cover back. The vehicle cover shall be rolled back as follows:

a. Unfasten apron of cover from brackets 10919932 and 10919933 back to third bow of frame from front.

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b. Open clamps attached to those brackets from which cover apron has been unfastened and remove freed tie rods.
c. Open sliding fastener at front of cover and roll cover back to second rod.
d. Tuck cover apron between frame and deck on right side to clear exhaust openings.

3.3.26 Marking. In addition to any special marking required in the contract (see 6.2), the vehicle shall be marked in accordance with MIL-STD-129.

3.3.26.1 Lifting points. The information "LIFT HERE" shall be stenciled adjacent to the lifting eyes with arrows pointing to the lifting eyes, using white enamel conforming to TT-E-529. Stenciling shall be in characters at least 1/2 inch high.

3.3.26.2 Closure marking. The information "TO LIFT VEHICLE FOR LOADING, UNFASTEN AND ROLL COVER FROM FRONT AND REAR BOWS EXPOSING LIFTING EYES," shall be stenciled in a conspicuous location on the front and rear of the vehicle closure, using lacquer conforming to MIL-L-14486, except that the color shall be white. Stenciling shall be in characters at least 3/4 inch high. The information "CLOSURE KIT, VEHICLE PROTECTIVE - M578 - DO NOT DESTROY - REINSTALL FOR SHIPMENT OR STORAGE OR SHIP WITH OVE" shall be stenciled in a conspicuous location on one end and one side of the vehicle closure, using lacquer conforming to MIL-L-14486, except that the color shall be white. Stenciling shall be in characters at least 2 inches high.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection records. The contractor shall maintain records of all inspections performed and such records shall be readily available for review by the Government representative.

4.2 First article inspection. One of the first 10 production processed vehicles (see 3.1) shall be subjected to the inspections specified in 4.5.

4.3 Inspection of production processed vehicles. Unless otherwise specified (see 6.2), all production vehicles shall be subjected to the inspections specified in 4.5.2 through 4.5.2.3.

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4.4 Rejection. Failure of the first article, or any production vehicle, to conform to the applicable requirements of this specification, shall be cause for rejection of the vehicle by the Government. The vehicle shall not be accepted until objective evidence that the contractor has corrected the condition causing the rejection has been provided to the Government.

4.5 Methods of inspection.

4.5.1 Materials. To determine conformance to 3.2, all materials to be used in processing of vehicles shall be inspected in accordance with material specifications. Certified inspection and laboratory test reports shall be provided which show that furnished materials conform to the applicable material specifications, except for materials that have been inspected by the Government at source. Materials listed on a Qualified Products List shall be obtained from one of the approved sources indicated.

4.5.2 Processing. Vehicle processing shall be inspected to determine conformance to this specification. Inspection of processing shall include all requirements specified in table I and inspections 4.5.2.1 through 4.5.2.3.

4.5.2.1 Cleaning. The interiors of the vehicles shall be examined for cleanliness to determine conformance to 3.3.4.1. One vehicle each day shall be tested for cleanliness in accordance with the applicable provisions of MIL-P-116. To determine conformance to 3.3.4.2, the exterior of the vehicle shall be examined for cleanliness. The surfaces on which tape is to be applied, shall be examined for cleanliness before application.

4.5.2.2 Fuel tank. To determine conformance to 3.3.5.5.6, visual inspection of preservative application shall be accomplished to assure addition of the correct amount of oil based upon a known amount of residual fuel.

4.5.2.3 Cooling system. To determine conformance to 3.3.6, one processed vehicle shall be selected at random from each day's production for inspection. The engine coolant shall be tested using a hydrometer-thermometer type tester with a range of minus 60°F to plus 160°F conforming to GG-T-250.

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TABLE I. Processing inspection.
(see indicated paragraphs for level A, and B, requirements)

	Cleaning	Preservation		Packaging
	Levels A and B	Level A	Level B	Levels A and B
Preparation prior to processing				3.3.1
Processing records				3.3.1.1
Disassembly				3.3.2
Record forms				3.3.3
Cleaning and drying	3.3.4			
Vehicle interior	3.3.4.1			
Battery supports and retainers	3.3.4.1.1			
Backrests, seats, and crash pads	3.3.4.1.2			
Fuel filter cases	3.3.4.1.3			
Vehicle exterior	3.3.4.2			
Relubrication		3.3.5.1	3.3.5.1	
Preservation of battery supports and retainers		3.3.5.2	3.3.5.2	
Transmission, final drives and auxiliary drive gearbox*		3.3.5.3	3.3.5.3	
Engine crankcase*		3.3.5.4	3.3.5.4	
Engine preservation*		3.3.5.5	3.3.5.5.1	
Preservation through fuel system		3.3.5.5.2		
Preservation through combustion chamber		3.3.5.5.3		
Preservation through air intake and exhaust systems		3.3.5.5.4	3.3.5.5.4	
Preservation through dipstick shroud opening		3.3.5.5.5	3.3.5.5.5	
Fuel tank		3.3.5.6	3.3.5.6.1	
Cooling system*		3.3.6	3.3.6	
Water and antifreeze procedure		3.3.6.1	3.3.6.1	
Water and corrosion inhibitor procedure		3.3.6.2	3.3.6.2	
Antifreeze compound procedure		3.3.6.3	3.3.6.3	
Dry charge batteries and cables		3.3.7.1	3.3.7.1	
Electrolyte		3.3.7.2	3.3.7.2	3.3.7.2.1 and 3.3.7.2.2
Traverse ring gear		3.3.8	3.3.8	
Spade		3.3.9	3.3.9	
Winch and derrick assembly		3.3.10	3.3.10	
Machinegun mount		3.3.11	3.3.11	

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TABLE I. Processing inspection - Continued.
(see indicated paragraphs for level A, and B, requirements)

	Cleaning	Preservation		Packaging
	Levels A and B	Level A	Level B	Levels A and B
Miscellaneous securement		3.3.12	3.3.12	
Vision blocks				3.3.13
Cloth items				3.3.14
Hydraulic systems		3.3.15	3.3.15	
Backrests, seats, and crash pads				3.3.16
Hatches and doors		3.3.17	3.3.17.1	
Fire extinguishers				3.3.18
Miscellaneous preservation		3.3.19	3.3.19	
Basic issue item	3.3.20	3.3.20	3.3.20	3.3.20
Vehicle closure				3.3.21
Ventilation				3.3.21.1
Screens				3.3.21.1.1
BII containers				3.3.22
BII rack				3.3.22.2.1
Tow hooks				3.3.23
Loading				3.3.24
Reprocessing after loading		3.3.25	3.3.25	
Marking				3.3.26
Lifting points				3.3.26.1
Closure marking				3.3.26.2

* Inspect DD Form 1397

5. PACKAGING

This section is not applicable to this specification.

6. NOTES

6.1 Intended use. This specification covers processing of the 8736340 vehicle for storage outside of buildings, immediate use, domestic or overseas and including carloading.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable level of processing (see 1.2).
c. If first article inspection is not required (see 3.1).
d. If draining of residual fuel from fuel tank is required for Level B processing (see 3.3.5.6.1).
e. Applicable procedure for protection of cooling system (see 3.3.6).
f. If installation of BII containers on the railroad car shall be other than as specified (see 3.3.22.1).

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g. If special marking is required (see 3.3.26).
h. If inspection shall be other than as specified (see 4.3).

6.3 Safety precautions. Caution should be exercised in handling carbon dioxide (CO_2) fire extinguisher cylinders. Cylinders should not be dropped, permitted to strike each other, or be handled roughly. Extreme care should be exercised during the reinstallation operation to avoid tripping the fire extinguisher control trigger (see 3.3.18).

6.4 Forms. A copy of the "Equipment Log Book" and all required forms (see 3.3.3) will be furnished to the contractor by the Government at least 30 days before shipment of the vehicles required by the contract delivery schedule.

6.5 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged.

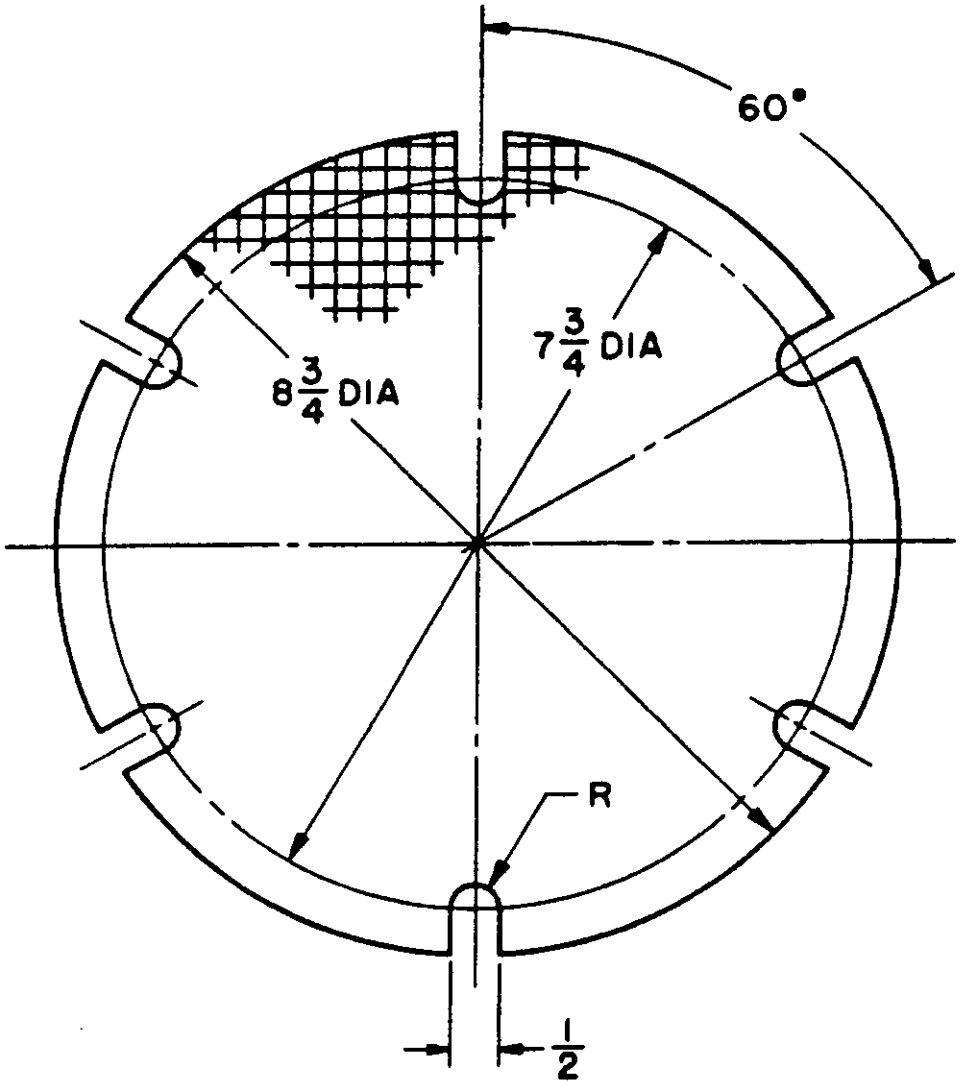
6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project No. PACK-A302)

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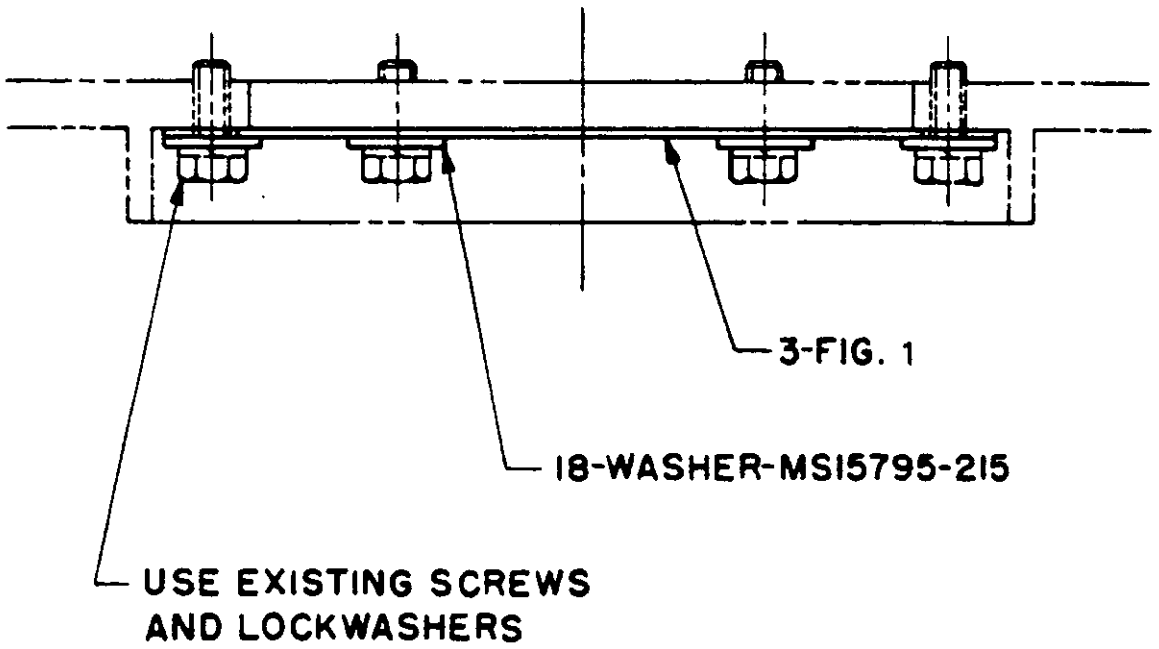
MATERIAL

CLOTH, WIRE
TYPE I, CLASS 1
STEEL, LOW CARBON
HOT GALVANIZED AFTER WEAVING
SPEC. RR-W-360
4 X 4 MESH, .047 DIA. WIRE

TOLERANCE - ± 1/16

FIGURE 1. Screen

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APPENDIX

REQUIREMENTS FOR ASSEMBLY, INSTALLATION, AND REMOVAL OF VEHICLE CLOSURE FOR M578 VEHICLE

10. SCOPE

10.1 This appendix covers requirements for assembly, installation, and removal of the vehicle closure (see 3.3.21).

20. APPLICABLE DOCUMENTS. None

30. REQUIREMENTS

30.1 Assembly and installation of closure. The vehicle closure 10914995 shall consist of a closure kit frame 10914996 and a cover 10914997 and subordinate drawings. The cover shall be attached after the frame has been installed on the vehicle in accordance with hull modification 10906906, and 10906907 and subordinate drawings.

30.1.1 Frame installation. The closure frame shall be assembled and installed in accordance with the following instructions:

a. Attach struts 10908831 and 10908834 to hull using screws and lockwashers.
b. Attach brackets 10908836, 10908837-1, and 10908837-2 to hull using screws and lockwashers.
c. Attach brackets 10919924-1 and 10919924-2 to hull using screws and nuts.
d. Fasten base frame angles 10908551 and 10908552 to struts 10908834 using washers 10908548, screws and nuts.
e. Fasten base frame angles 10908551, 10908552, 10908561, 10915005-1, and 10915005-2 to brackets and to each other using angles 10908556, screws, and nuts.
f. Fasten front base frame angles 10908830-1 and 10908830-2 together and to strut 10908831 using angle 10908556, screws and nuts.
g. Fasten rear base frame angles 10908830-1 and 10908830-2 together using angle 10908556, bracket 10915006, screws, and nuts.
h. Fasten front and rear base frame angles to side base frame angles to form corner joints using screws and nuts.
i. Using 7/8 nuts and lockwashers, install brackets 10914998 and posts 10908550 on side base frame angles. Install outer brackets 10914998 on the front and rear of base frame angles using 7/8 screws and lockwashers.
j. Place bows 10908853 over posts 10908550 along each side of base frame. Slip tubes 10908828-2 onto connectors of tubes 10908848 and place this assembly over posts 10908550 at front and rear base frame.

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APPENDIX

REQUIREMENTS FOR ASSEMBLY, INSTALLATION, AND REMOVAL OF VEHICLE CLOSURE FOR M578 VEHICLE

k. Place links 10908557 and 10908558 over connectors of bows 10908853. Slip bows 10915007 and 10915008 over connectors of bows 10908853. Place remaining links 10908557 and 10908558 over connectors of bows 10915007. Slip plates of tubes 10908848 over connectors of bows 10915007 at front and rear. Insert connectors of bows 10915007 into opposing bows 10915008.
l. Install braces 10908835 using washers 10908548, screws, nuts, and clamps.

30.1.2 Installation of cover. The vehicle cover shall be installed over the closure frame in accordance with the following instructions:

a. Drape cover 10914997 over frame 10914996.
b. Slip rods 10908553 thru side loops and rods 10908555 thru front and rear loops in cover.
c. Attach studs 549198 to brackets 10919932 and 10919933 using screws with lockwashers attached.
d. Attach clamps 10922144 and brackets 10919933 to brackets 10914998 and strut 10908831 at front of frame using spacer 10919946 screws, nuts, and washers. Attach clamps 10922144 and brackets 10919932 to remainder of brackets 10914998 and to bracket 10915006 at rear of frame using spacers 10919946 screws, nuts, and washers.
e. Position clamps to obtain good fit of cover by means of slotted holes in brackets 10914998. Secure rods 10908553 and 10908555 to brackets with clamps.
f. Close both ends of cover with sliding fasteners. Lock fasteners with cotter pins 590205.
g. Secure apron of cover to brackets 10919932 and 10919933 using turn-button stud fasteners 549198.

30.2 Removal of vehicle closure. The vehicle cover and frame shall be removed from the vehicle in a step by step procedure that is the reverse of the installation procedure (see 30.1.1 and 30.1.2).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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